ML/DL INTERNSHIP · SPRING 2026 · ACIBADEM MAAÜ

Predicting Village Wealth from Space

A modern replication & fairness audit of Yeh et al. (2020), Nature Communications

Onur Haniffa · Advisor: Dr. Seda Nilgün Dumlu · 22 May 2026

MOTIVATION

The Problem & the Promise

THE PROBLEM

- Reliable poverty data is scarce, expensive, and infrequent
- Thinnest exactly where the need is greatest
- Decisions get made on stale or missing numbers

THE PROMISE

- Satellites image everywhere on Earth, cheaply and repeatedly
- Can a neural network read household wealth from imagery?
- If yes: a scalable complement to ground surveys

BACKGROUND

Where This Comes From

THE LINEAGE

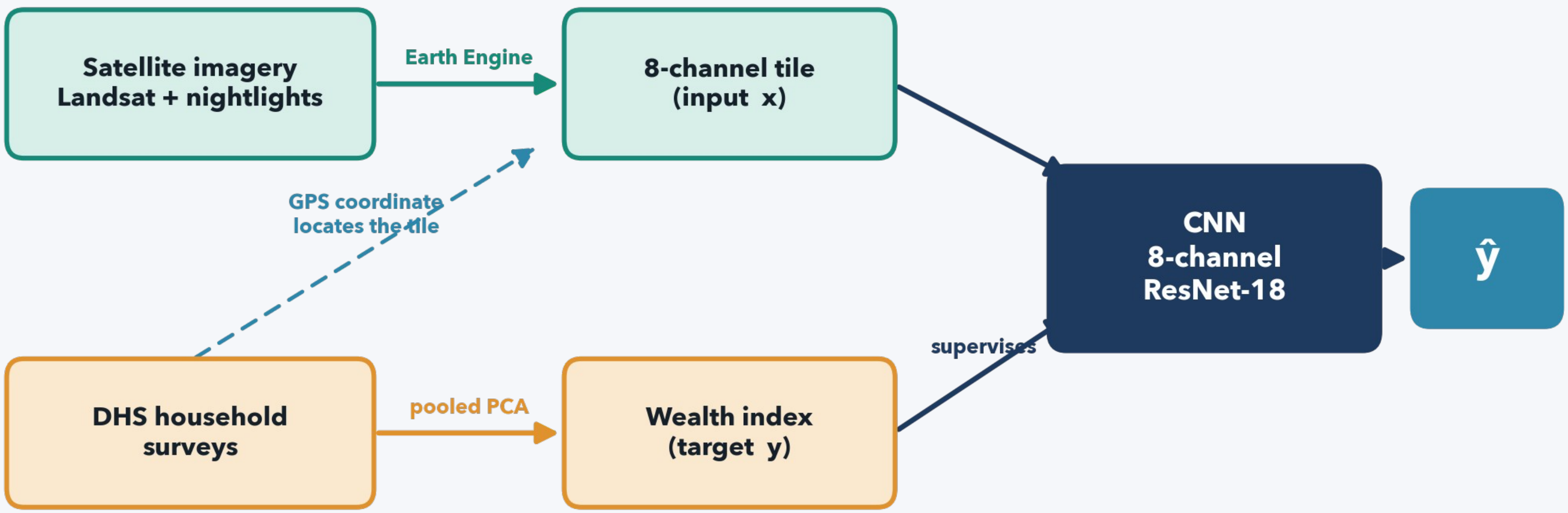
- Jean 2016 (Science): nightlights + transfer learning
- Yeh 2020 (Nature Comms): mean $r^2 = 0.70$ across 23 countries
- Aiken 2023 (IJCAI): fairness gaps – but only 10 countries

OUR FOUR CONTRIBUTIONS

- Replicate Yeh in modern PyTorch
- Fairness audit across ALL 23 countries
- Uncertainty-aware fairness (novel)
- Temporal fairness drift (novel)

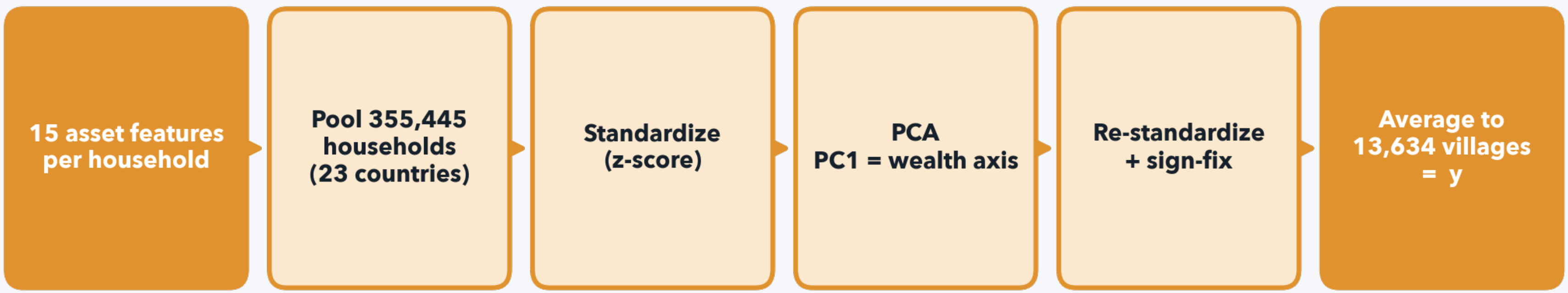
APPROACH

Two Data Sources, One Model



Each village's GPS point pulls a satellite tile (x); its survey gives a wealth value (y); the CNN learns x → y.

From a Survey to a Wealth Number



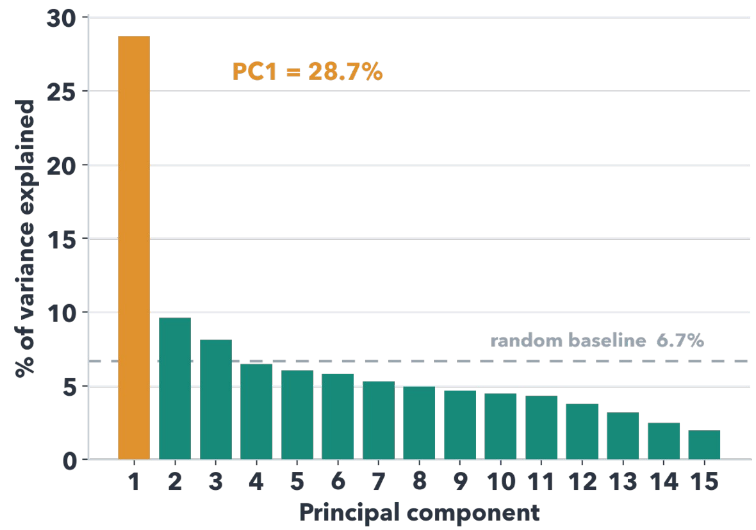
Assets, not income. Pool every household across 23 countries, run PCA, take PC1 as the wealth axis, average to the village.
PC1 explains 28.7% of total variance.

METHOD · VALIDITY

Is PC1 a Valid Wealth Index?

WHY 28% IS PLENTY

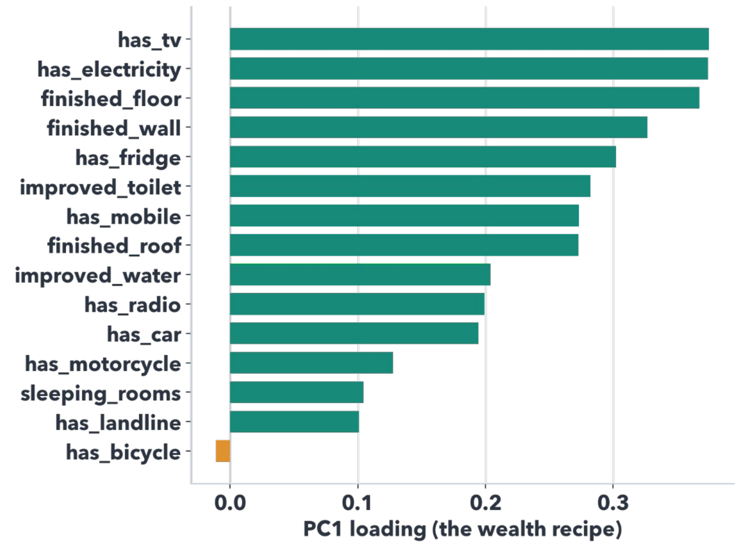
- PC1 = 28.7% – but the benchmark is the 6.7% random floor, not 50%
- That is 4.3× the floor: one strongly dominant factor
- Binary survey data caps variance; 20–40% is the norm
- Variance = compactness, not validity
- Validated by the country ranking + urban/rural gap



Why PC1 Is Wealth, Not PC2/PC3

LEVEL VS CONTRAST

- A wealth factor makes ALL assets rise together (a 'level')
- PC1 is all-positive → that IS wealth
- PC2/PC3 are orthogonal contrasts (mixed signs)
- PC2 = rural/traditional vs urban/modern; PC3 = big durables
- Wealth is the #1 driver → it lands on PC1 by construction



METHOD · IMAGERY

From a GPS Point to a Tile

DHS cluster GPS point (lat, lon) → Earth Engine centers a 6.72 km box → Landsat L8/L7/L5 + nightlights 3-year median → 8-channel tile 224×224 px @ 30 m → linked by cluster_id to wealth y

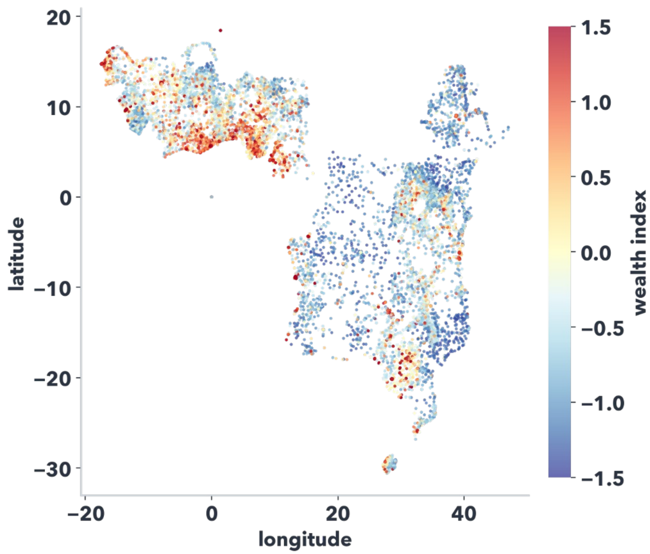
RGB · NIR · SWIR1 · SWIR2 · thermal · nightlights

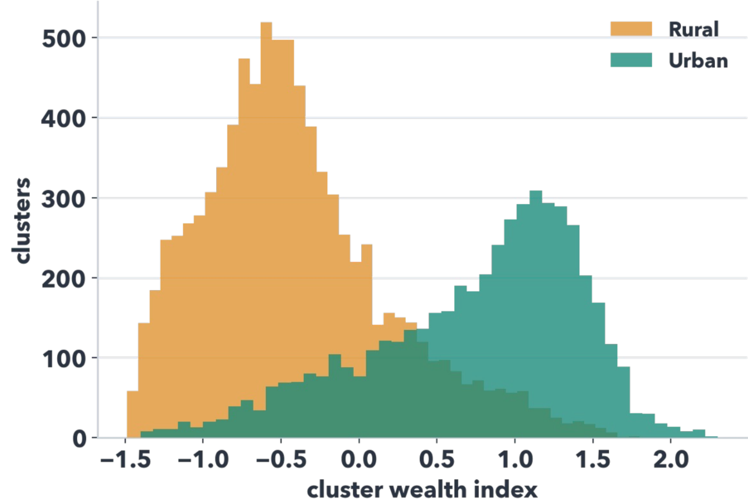
The GPS coordinate selects the patch; Earth Engine builds a cloud-free 3-year median; 8 bands stack into one 224×224 tile.

Decisions & Honest Divergences

- Ordinal categoricals (floor/wall/roof, water, toilet) collapsed to a binary 'improved/finished' flag – documented divergence
- Quality filter: drop households missing >30% of features – only ~30 of 355,445 dropped (99.99% complete)
- Special DHS codes handled (e.g. 'rooms' 96/97/98/99 → missing, not a fake 10-room house)
- Nightlights: DMSP-OLS (pre-2012) vs VIIRS (2012+) – different scales, documented
- One survey round per country → 13,634 clusters (Yeh pooled multiple rounds → 19,669)

What the Data Looks Like





13,634
georeferenced clusters

28.7%
PC1 variance

1.16σ
urban/rural gap

STATUS

Status & Plan

DONE

- DHS → wealth index, all 23 countries
- 355,445 households → 13,634 villages
- Full EDA

IN PROGRESS

- Satellite tiles via Earth Engine
- ~50% extracted
- Finishing in ~1–2 days

NEXT

- Pair tiles ↔ wealth
- Train 8-channel ResNet-18
- Replication r^2 → fairness → extensions

Target mean-of-folds $r^2 \geq 0.60$ • **Deadline** 30 May 2026